A2.3 Viruses

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a2-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 88–99 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

What a virus is

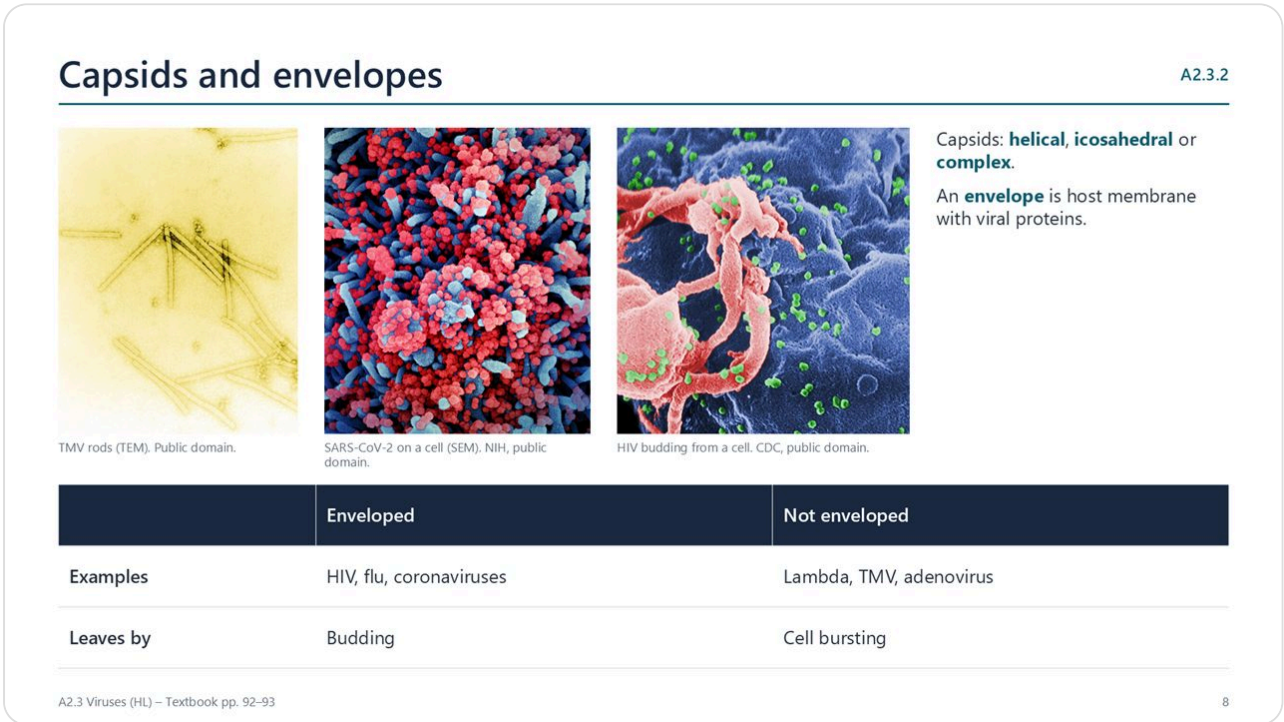
A2.3.1 Features shared by all viruses

pp. 88–90

Viruses are very small, from about 20 to 400 nm, and of a fixed size, because they are assembled from parts rather than growing. Each has **DNA or RNA**, never both, inside a **capsid** of protein subunits. They have **no cytoplasm**, no ribosomes and few or no enzymes, so they can reproduce only inside a host cell, using its enzymes and ribosomes. A virus is not a cell.

A2.3.2 Diversity of viruses

pp. 90–93

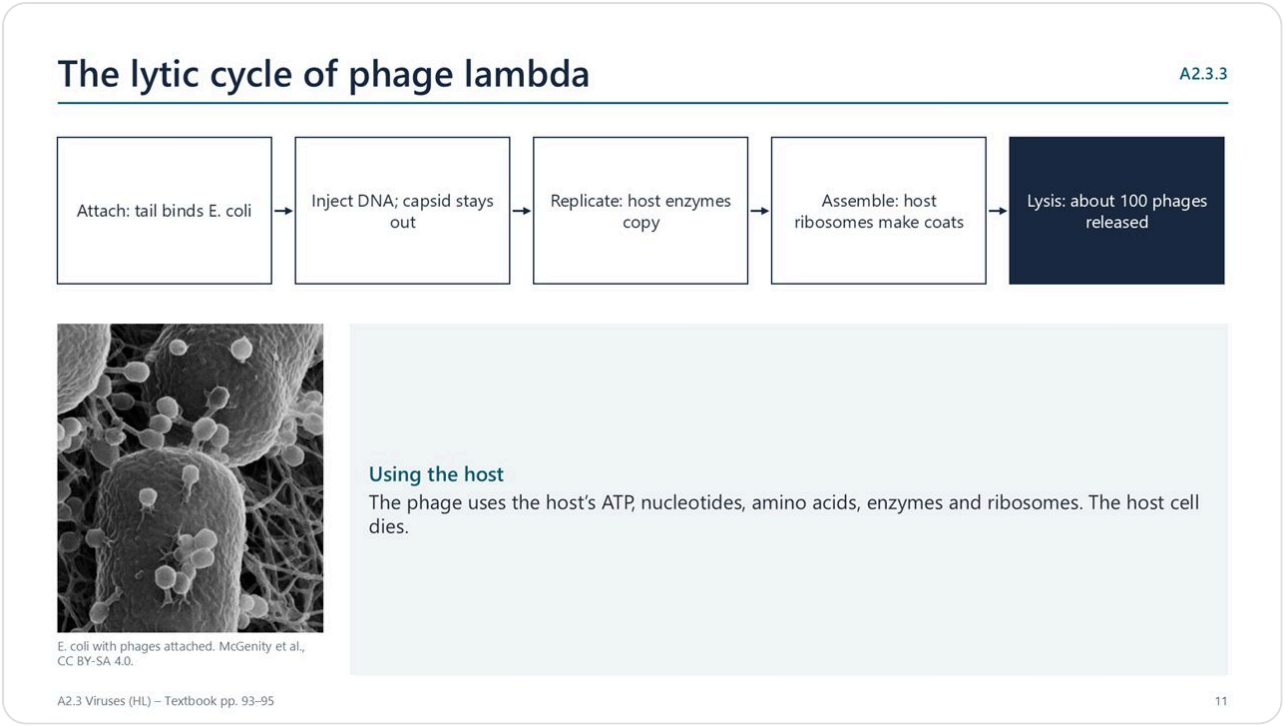


Genomes vary: double-stranded DNA in **phage lambda**, positive-sense single-stranded RNA in **coronaviruses**, which acts directly as mRNA, and two copies of single-stranded RNA in **HIV**, a retrovirus whose **reverse transcriptase** makes DNA from it. Capsids may be helical, icosahedral or complex. Some viruses, such as HIV and coronaviruses, have an **envelope** of host membrane with viral glycoproteins, taken as they bud out of the cell. The great variety suggests viruses did not all evolve from one ancestor.

Reproduction

A2.3.3 The lytic cycle

pp. 93–95



Phage lambda attaches to a receptor on *E. coli* and injects its DNA. The host's enzymes replicate the phage DNA and its ribosomes make capsid proteins, using the host's ATP, nucleotides and amino acids. New phages assemble, and enzymes break down the cell wall, so the cell bursts (**lysis**) and releases about a hundred phages.

A2.3.4 The lysogenic cycle

pp. 95–96

Instead of taking over the cell at once, lambda DNA can be inserted into the bacterial chromosome as a **prophage**. It is copied every time the bacterium divides, and the cell survives. Stress, such as UV light, can cut the prophage out, and the lytic cycle begins. Lysogeny is favoured when bacteria grow poorly, because lysis would then release few phages.

Origins and evolution

A2.3.5 Where viruses came from

pp. 96–98

Three hypotheses compete: **progressive** (genes that escaped from cells, such as transposons, gained a coat), **regressive** (cells that became parasites and lost most of their genes) and **virus first** (self-replicating RNA before cells). No gene is shared by all viruses, so they probably arose several times. Their shared features, such as capsids and small size, are best explained by **convergent evolution** to a parasitic life. They use the same genetic code as their hosts.

A2.3.6 Rapid evolution: influenza and HIV

pp. 98–99

Viruses have short generation times, huge numbers and, in RNA viruses, no proofreading, so mutations are frequent and selection by immunity or drugs is strong. In influenza, **antigenic drift** slowly changes the H and N surface antigens, so vaccines are updated each year; **antigenic shift**, the swapping of whole RNA segments between strains, can produce a new subtype and a pandemic. HIV's reverse transcriptase makes many errors, so many variants arise in one person; combinations of drugs are used so that resistance is unlikely.

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